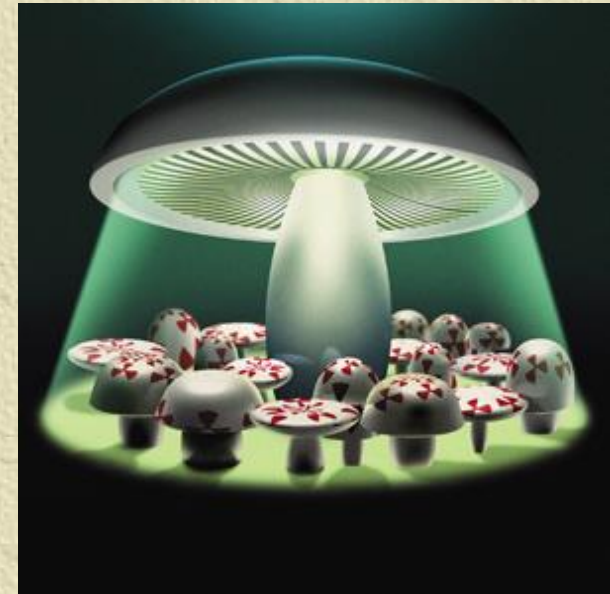

Properties, structure & classification of fungi



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Objectives:

- ❖ To clarify the properties ,advantages and classification of medically important fungi with their clinical significances as etiologic factors in human diseases .



Introduction

- ❖ The study of fungi is Mycology and fungal infections are Mycosis.
- ❖ The term "mycology" is derived from Greek word "**mykes**" meaning mushroom Therefore mycology is the study of fungi.
- ❖ Approximately 400.000 species of fungi are found in water, soil, organic debris and vegetables.



✧ Fungi were discovered earlier than bacteria and viruses.

- ✧ • In the past, most fungi cause skin infections or cosmetic infections, where bacteria and viruses cause serious fatal diseases, so there was no interest of studying fungi.
- ✧ • In 1980, when HIV infection was discovered, increasing number of immunocompromizing conditions, they found that fungi produce fatal diseases; from that time, fungi return to be in focus again.



ECONOMIC IMPORTANCE OF FUNGI

25-50% of harvested fruits and vegetables are damaged by fungi.

Fungal infections of plants are commonly called rots, rusts, blights & wilts,

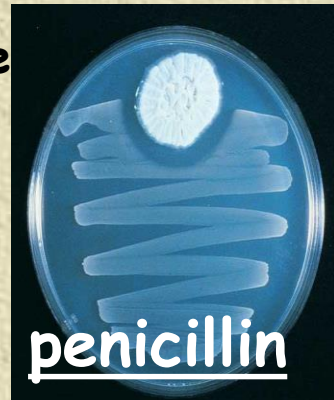
Phytophthora infestans: Caused great potato famine in mid-1800s. Over 1 million people died from starvation in Ireland. Many immigrated to the U.S.



Beneficial fungi:

Candida oleophila: Prevents fungal growth on harvested fruits.

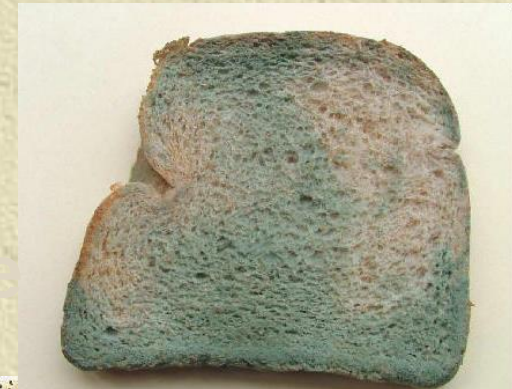
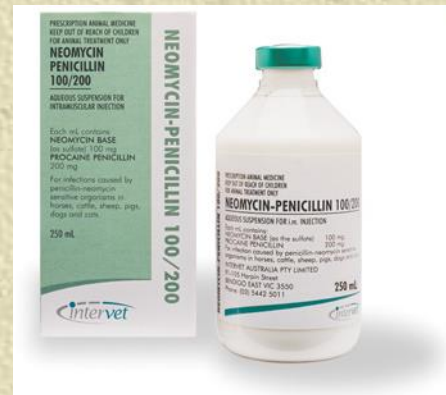
- ♦ ***Saccharomyces cerevisiae***: Used to make bread and wine.
- ♦ **Genetically engineered yeast strains** are used to make (Hepatitis B vaccine).
- ♦ ***Taxomyces***: Produces anticancer drug taxol.
- ♦ ***Trichoderma***: Produces cellulase. Used to make fruit juice.



Food

Where

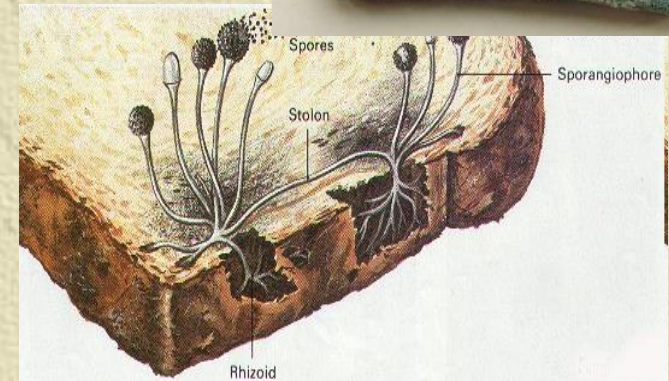
Vaccines for Hepatitis B – *Sacchromyces cerevisiae*



Food



Storage



Factors predisposing to fungal infections:

- Long administration of broad spectrum antibiotics.
- Immunological status of the patient.
- 10% of the AIDS patients develop mycoses.
- cancer patients
- Age
- Surgical procedures
- Immunosuppressive drugs
- Irradiation therapy
- Indwelling catheters & prosthetic devices: catheter, heart valves.
- Obesity
- Drug addiction
- Transplants
- Occupation
- Organ damage (especially chronic) heart diseases, diabetics, renal failures, third grade burns, malnutrition, alcoholism, drug abuse,...



Comparison between fungi & bacteria

Feature	Fungi	Bacteria
Diameter	Approximately $4\mu\text{m}$ (conidia)	Approximately $1\mu\text{m}$ (staph)
Nucleus	Eukaryotic	Prokaryotic
Cytoplasm	Mitochondria& endoplasmic reticulum present	Mitochondria & endoplasmic reticulum absent
Cell membrane	Sterols present	Absent(except <i>Mycoplasma</i>)
Cell wall content	chitin	peptidoglycan
Spores	Sexual & asexual spores for reproduction	Endospores for survival ,not for reproduction
Dimorphism	Yes(some)	No
Metabolism	Require organic carbon ,no obligate anaerobes	Many do not require organic carbon many obligate anaerobes

Medical Mycology



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- ✦ It is the science that deals with the study of pathogenic fungi that produce diseases.
 - ✦ Structure
 - ✦ Fungi are eukaryotic organisms have true nuclei with definite nuclear membrane, nucleolus, cytoplasmic organelles.
 - ✦ Cell membrane of fungi has sterols, which is the target of action of antifungal agents.
 - ✦ Ergosterol dominates in contrast to cholesterol in mammalian membrane.

Structure

✧ Cell wall of fungi lacks:

- ✧ - Peptidoglycan - Glycerol & ribitol teichoic acid - Lipopolysaccharide

✧ Cell wall composed of:

1-Structural components:

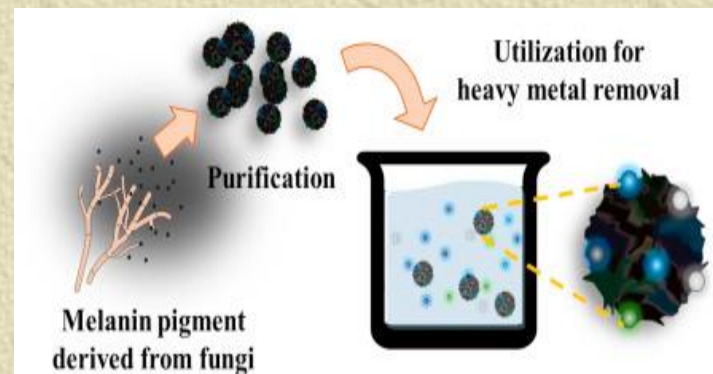
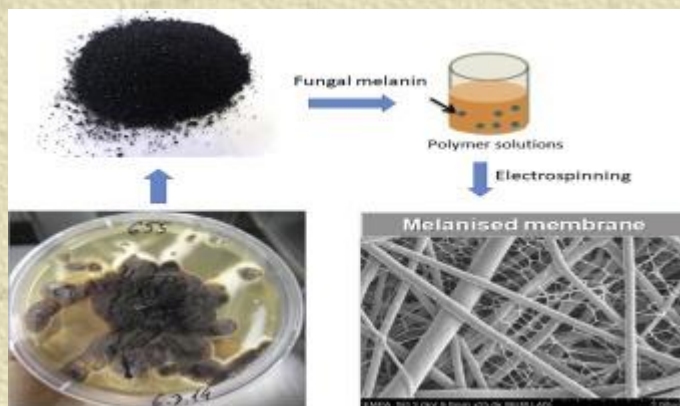
- ✧ chitin microfibrils [$\beta(1-4)$ -linked polymer of N-acetylglucosamine]
- ✧ chitosan in Zygomycota [$\beta(1-4)$ -linked polymer of glucosamine]
- ✧ β -linked glucans

2-Gel-like components:

- ✧ Mannoproteins (form matrix throughout wall)

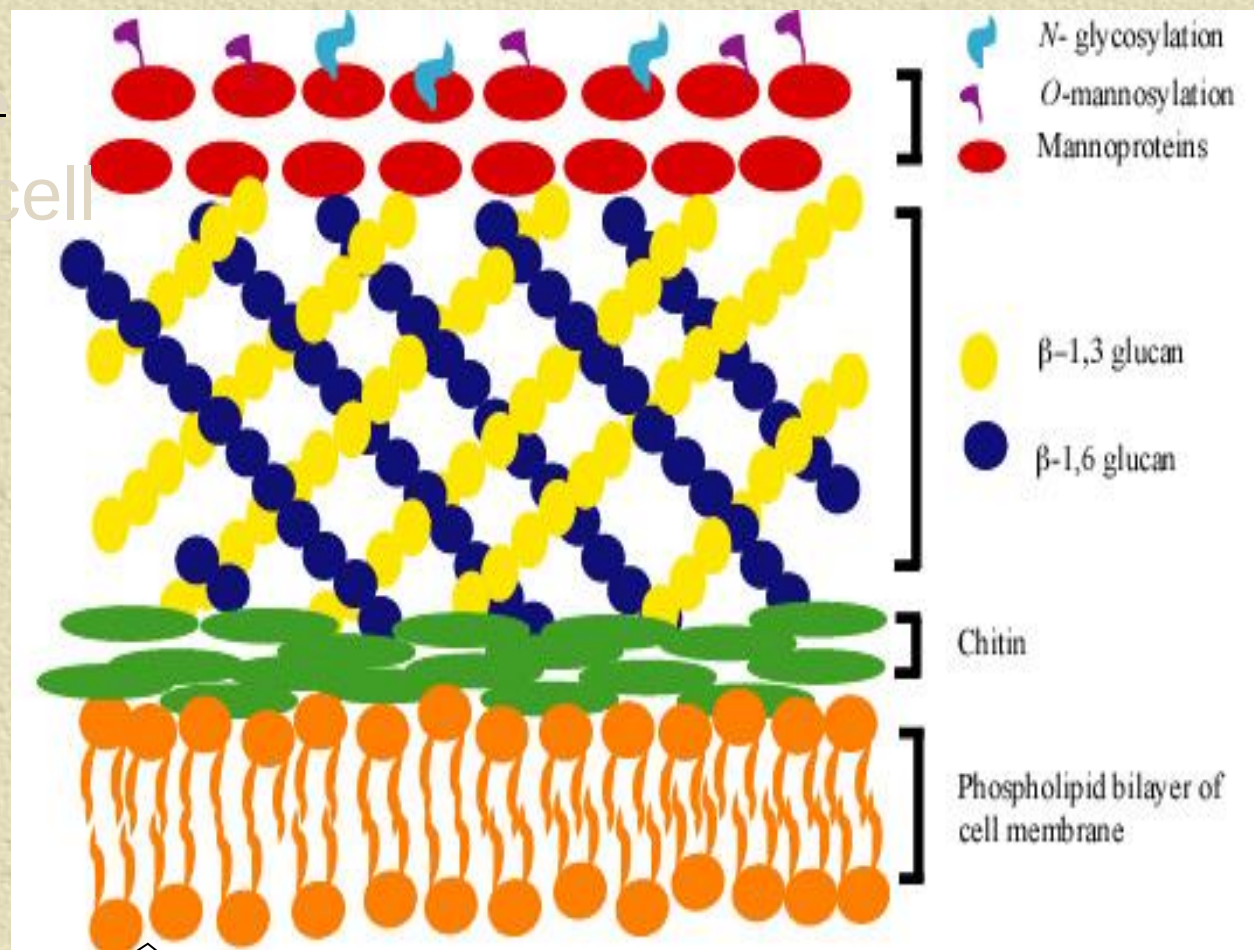
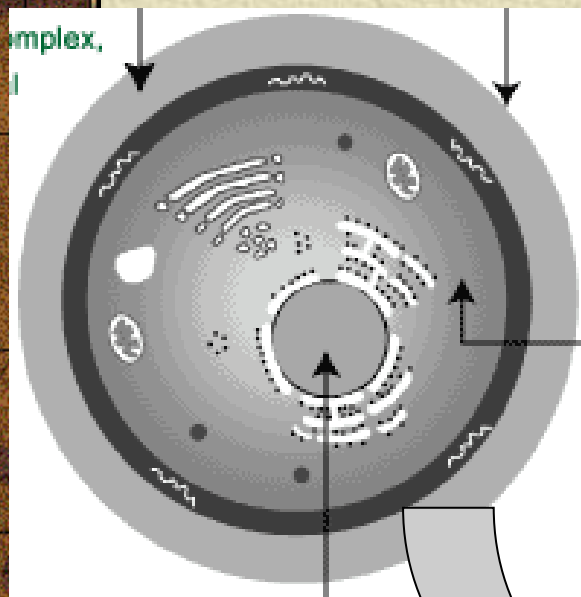
Other cell wall components

- ✧ Antigenic glycoproteins, agglutinins, adhesions—on cell wall surface
- ✧ **Melanins**—dark brown to black pigments (confer resistance to enzyme lysis, confer mechanical strength and protect cells from UV light, solar radiation and desiccation)



The Fungal Cell Wall

Fluid mosaic
 surrounding the cell



Fungal metabolism:

- ✦ All fungi are heterotrophic organism need to be parasite or saprophyte (on plant, animal or human) to obtain organic source of carbon or nitrogen.
- ✦ All fungi are aerobic.
- ✦ Some are facultative anaerobes.
- ✦ None are strict anaerobes.

Fungal metabolism continue

- ✦ Optimum temperature of growth 25 - 30 °C (because most are saprophyte live in the environment).
- ✦ Some are thermophilic, and can grow at higher temperature.
- ✦ Fungi can tolerate a wide range of pH (2 - 9), but generally they prefer acidic media.

NUTRITIONAL ADAPTATIONS OF FUNGI

Fungi absorb their food, rather than ingesting it.

- Fungi grow better at a pH of 5, which is too acidic for most bacteria.
- Almost all molds are aerobic. Most yeasts are facultative anaerobes.
- Fungi are more resistant to high osmotic pressure than bacteria.
- Fungi require less nitrogen than bacteria to grow.
- Fungi can break down complex carbohydrates (wood, paper), that most bacteria cannot.



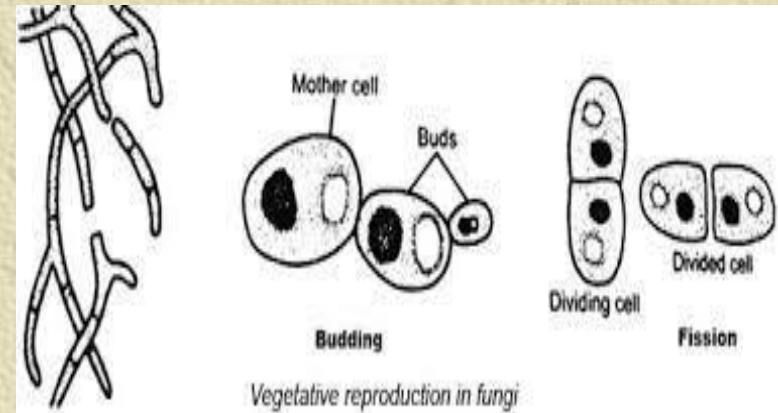
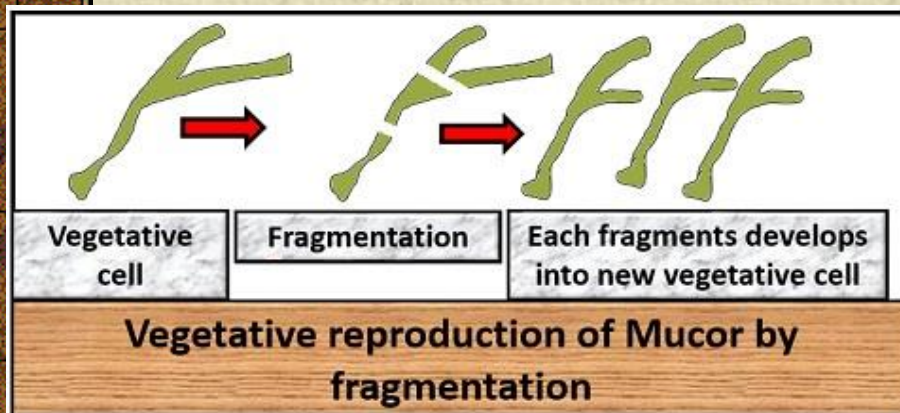
Reproduction

1-Sexual reproduction: Involve the union of 2 nuclei or 2 sex cells or 2 sex organs.

2-Asexual reproduction:

It is the main method of reproduction, it includes:

- **Fragmentation** of hyphae & each fragment grows into a new individual fungus.
- **Fission** of cell into 2 daughter cells (similar to binary fission in bacteria).
- **Budding** of cells, each bud produce new individual (e.g. *Candida*).
- Formation of asexual spores.



✧ A single fungus may contain both modes of reproduction (sexual and/or asexual).

✧ **Fungal Spores**

✧ It is the method of reproduction in fungi unlike bacteria.

✧ Two types:

✧ 1-Sexual

✧ 2-Asexual

Fungal Spores

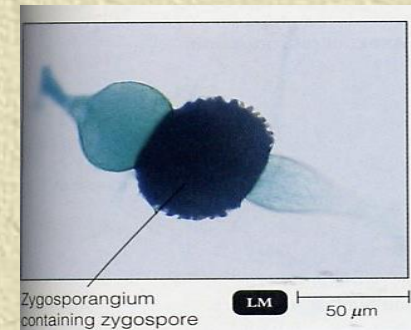
✦ Sexual spores - exhibit fusion of nuclei.



1 - Ascospore - spore formed in a sac-like cell known as an ascus. Often eight (8) spores formed. (Ascomycetes)

2 - Zygospore - a thick-walled spore formed during sexual reproduction (Zygomycetes)

3 - Basidiospore - sexual spore produced on a specialized club-shaped structure, called a basidium. (Basidiomycetes)



Asexual spores - most common type.

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- 1 - Conidia .
 - 2 - Arthroconidium (Arthrospore) .
 - 3 - Blastoconidia (Blastospore).
 - 4 - Chlamydospore.
 - 5 - Sporangiospore.
 - 6 - Thallospore .

Immunity to fungal infections:

- ✧ Mechanism of immunity to fungal infections can be innate or acquired.

- ✧ The non-specific immunity includes the physical barriers offered by skin and mucus membranes along with their secretions and normal flora. The pH, body temperature and serum factors along with phagocytic cells play an important part in providing non-specific immunity.

Cell mediated immunity:

- ✧ Immunity is provided **non-specifically** by effector cells (polymorphonuclear leucocytes) and professional phagocytes (monocytes and macrophages) and **specifically** by T lymphocytes. The phagocytes are very important in defence against **Candida, Aspergillus** and **Zygomycetes**
- ✧ Expression of T-cell-mediated immunity to fungi includes:
 - delayed-type hypersensitivity
 - contact allergy
 - chronic granulomatous reactions

Humoral immunity

antibodies are produced against many fungi.

antibodies help in clearing fungal pathogens through opsonisation, which is important against *Candida* and *Cryptococcus*.

Another component of humoral immunity is the complement, which can act as opsonins and may even cause damage to their cells through complement activation.

Antibodies are important to fungal serodiagnosis.



Thanks a lot